Kyle

Game Developer

Portfolio: https://sadoe3.github.io/

PROFILE SUMMARY

C++ Game Programmer with a comprehensive understanding of both low-level systems and modern engine architecture. Built a solid foundation in memory management and multithreading through OS development (Pintos). Deepened expertise in core system mechanics by developing a custom physics engine and visual editor from scratch using Win32 and DirectX 12. Currently leveraging this C++ architectural expertise to master Unreal Engine's Gameplay Ability System (GAS) for scalable gameplay logic. Seeking a junior gameplay/engine role to apply system-level problem-solving and collaborate effectively with cross-functional teams.

PROJECTS

Custom Game Physics Engine (Individual Project)

Tech Stack: C++, DirectX 12

- Implemented a from-scratch impulse-based physics engine in C++ and DirectX 12 with iterative constraint solver and SAP broad-phase.
- Validated stability and performance under high-load stress, demonstrating complex interactions (chains, ragdolls, stacking) in sandbox scenes.
- Engineered an in-engine visual debugger featuring object picking, frame-by-frame rewinding, wireframe collision overlay, and real-time contact point/normal visualization.

Pintos Operating System Development (Individual Project)

Tech Stack: C

- Completed all four Stanford Pintos OS projects, implementing threading, virtual memory, and file systems entirely in C.

Dark Souls-like Action RPG Prototype (Team Project)

Tech Stack: Unity, C#

- Developed core combat systems and player interaction in a collaborative environment, leveraging Git and Slack for efficient version control and team communication.

PROFESSIONAL EXPERIENCE

Interface Developer | Siwon IT Co., Ltd

Jan 2026 – **Present**

- Spearheaded the development of an inter-hospital clinical information exchange system as the sole developer, modernizing the legacy XAVE API and integrating the HIRA API to meet complex business requirements.
- Resolved critical infrastructure and security bottlenecks by engineering a custom C++ middleware with asynchronous I/O and atomic writes, while architecting a session-isolated, large-scale data transfer pipeline utilizing IIS Reverse Proxy.

Application Developer | Jeonbuk National University Hospital

Mar 2025 – Aug 2025

- Collaborated with medical staff to integrate Antimicrobial Stewardship Program (ASP) protocols into the EMR system, developing modules to regulate antibiotic usage.
- Resolved UI freezing by architecting a multiprocessing loading system using Win32 API and Named Pipes, overcoming framework threading constraints.

EDUCATION

Bachelor of Science in Computer Science

- Academic Credit Bank System, South Korea
- Feb 2022 | 4.18 / 4.5

Advanced Diploma Program in Computer Programmer Analyst

- Fanshawe College, London, ON, Canada
- 2019 – 2020 | 4.06 / 4.2 (Dean's Honour Roll : Fall 2019, Winter 2020)
- Completed 3 of 6 semesters; paused due to COVID-19.